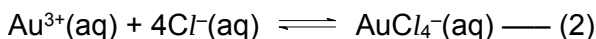
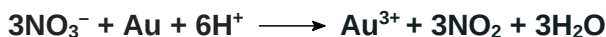


1 (a) (i) $E_{\text{cell}}^{\ominus} = 0.00 - (+1.50)$ [1]
 $= -1.50 \text{ V} < 0$, hence reaction is not spontaneous.



Due to the formation of chloroaurate(III) anions (AuCl_4^-) in equilibrium (2), the Au^{3+} ions are constantly removed from solution, keeping $[\text{Au}^{3+}]$ low causing position of equilibrium for equilibrium (1) to shift right.

[1]: equations showing formation of AuCl_4^-

[1]: explanation

(b) Amount of gold = $\frac{4 \times 2.18 \times 10^6}{197.0}$ [1]
 $= 4.424 \times 10^4 \text{ mol}$

Amount of SO_2 needed = $4.424 \times 10^4 \times \frac{3}{2}$ [1]
 $= 6.636 \times 10^4 \text{ mol}$ ecf

Min. volume of SO_2 needed = $6.636 \times 10^4 \times 24.0$ [1]
 $= 1.59 \times 10^6 \text{ dm}^3$ ecf



In the presence of Cl^- ligands, the partially filled 3d orbitals of Au are split into two levels (non-degenerate) with a small energy gap (d orbital splitting).

When light passes through, energy is absorbed and an electron in a lower energy d-orbital is promoted to a higher energy d-orbital. (d-d transition).

The colour of the solution is the complementary colour of the light absorbed.

$\text{Au}(\text{CN})_2^-$: d^{10} configuration / fully filled d subshell

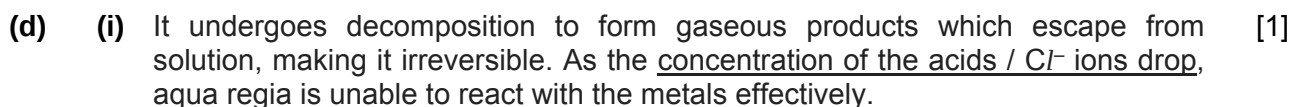
has energy gap but no vacant higher E d orbital for d-d transition.

[1]: partially filled 3d-orbital for AuCl_4^- ; small energy gap

[1]: d-d transition

[1]: fully filled 3d-orbital for $\text{Au}(\text{CN})_2^-$

[1]: no d-d transition



(ii) Refer to graph [2]

[1]: correct axes and labels: conc. of NOCl / molecule cm⁻³ vs time / s

[1]: correctly plotted points

(iii) Rate when conc. of NOCl = 1.5×10^{23} molecule cm⁻³ [2]

$$\begin{aligned}\text{Gradient of tangent} &= \frac{1.5 \times 10^{23}}{3.7 \times 10^3} \\ &= 4.05 \times 10^{19} \text{ molecule cm}^{-3} \text{ s}^{-1}\end{aligned}$$

Rate when conc. of NOCl = 0.75×10^{23} molecule cm⁻³

$$\begin{aligned}\text{Gradient of tangent} &= \frac{1.2 \times 10^{23}}{11.0 \times 10^3} \\ &= 1.09 \times 10^{19} \text{ molecule cm}^{-3} \text{ s}^{-1}\end{aligned}$$

When conc. of NOCl is doubled, rate is approximately quadrupled.

Rate of reaction is directly proportional to (conc. of NOCl)²

Order of reaction wrt NOCl is 2

[1]: find rate at 2 different points

[1]: conclude that rate $\propto [\text{NOCl}]^2$

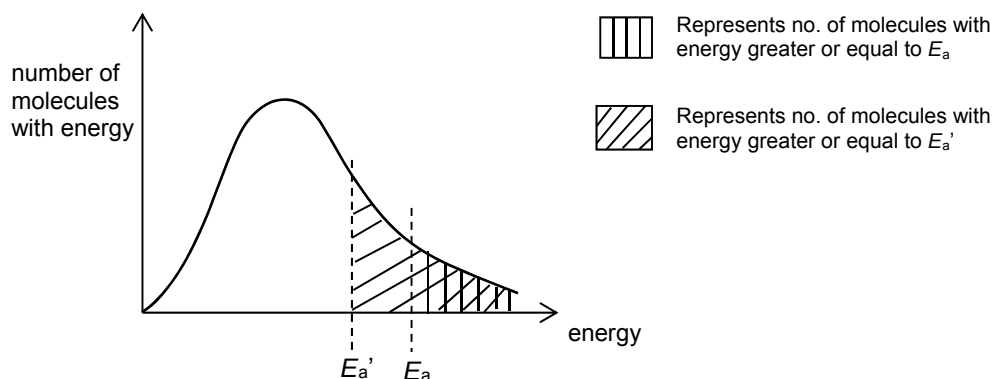
(iii) Rate = $k [\text{NOCl}]^2$ [2]

$$\begin{aligned}k &= \frac{\text{rate}}{[\text{NOCl}]^2} \\ &= \frac{4.05 \times 10^{19}}{(1.5 \times 10^{23})^2} \\ &= \frac{1.80 \times 10^{-27} \text{ molecule}^{-1} \text{ cm}^3 \text{ s}^{-1}}{1000} \\ &= \frac{1.80 \times 10^{-27}}{1000} \times 6.02 \times 10^{23} \\ k &= \underline{1.08 \times 10^{-6} \text{ mol}^{-1} \text{ dm}^3 \text{ s}^{-1}}\end{aligned}$$

[1]: k in terms of molecules⁻¹ cm³ s⁻¹ / find rate and conc. in terms of mol

[1]: k in terms of mol⁻¹ dm³ s⁻¹

(iv)



[2]

NO_2 speeds up the rate of decomposition by providing a different/alternative reaction path which has lower activation energy.

As shown on the diagram, the number of NOCI molecules with energy greater or equal to the lowered activation energy (E_a') will increase.

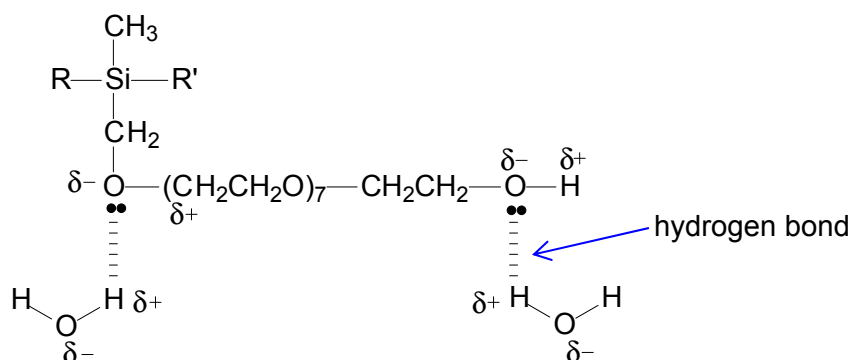
This results in an increase in the frequency of effective collisions.

Hence, the rate of decomposition increases.

[1]: correct axes, shape of graph (including beginning from origin) and correct shift of E_a

[1]: different reaction path with lower E_a and explanation (in terms of more molecules with energy $\geq E_a'$ and frequency of effective collision increase)

2 (a) (i) [1]



[1]: polarised O–H bond, lone pair of electrons on oxygen (can be on O of H₂O) and correct atoms involved in hydrogen bonding

Sufficient to show one site for hydrogen bonding

- (ii) The non-polar alkyl groups of molecule **B** forms weak instantaneous dipole–induced dipole attraction with the non-polar oil droplets. [2]
The polar groups of molecule **B** can form hydrogen bonds with / attracted to the water molecules.

Hence the dispersant can dissolve in both oil and water easily. This allows the oil to be broken down into smaller droplets and be removed more easily.

[1]: types of interactions molecule **B** form with oil and water
[1]: recognise that it can dissolve in both

(b) (i) giant molecular structure [1]

- (ii) From carbon to germanium, (less energy) is required to break the decreasing strength of covalent bonds between atoms due to the decreasing bond energy / increasing atomic radius of the atom down the group. [1]

- (iii) C is from period 2. It does not have energetically accessible d orbitals to accept lone pair of electrons from water molecules to form dative bonds. [1]
OR
The four large chlorine atoms shield the small carbon atom from the water molecules, making it difficult (high E_a) for the water nucleophile to attack the electrophilic C atom. (steric hindrance)

(accept explanation in terms of Si)

- (iv) $\text{GeCl}_4 + 2\text{H}_2\text{O} \longrightarrow \text{GeO}_2 + 4\text{HCl}$ [2]
 $\text{GeCl}_4 + 4\text{H}_2\text{O} \longrightarrow \text{GeO}_2 \cdot 2\text{H}_2\text{O} + 4\text{HCl}$
 $\text{GeCl}_4 + 2\text{H}_2\text{O} \longrightarrow \text{Ge(OH)}_4 + 4\text{HCl}$

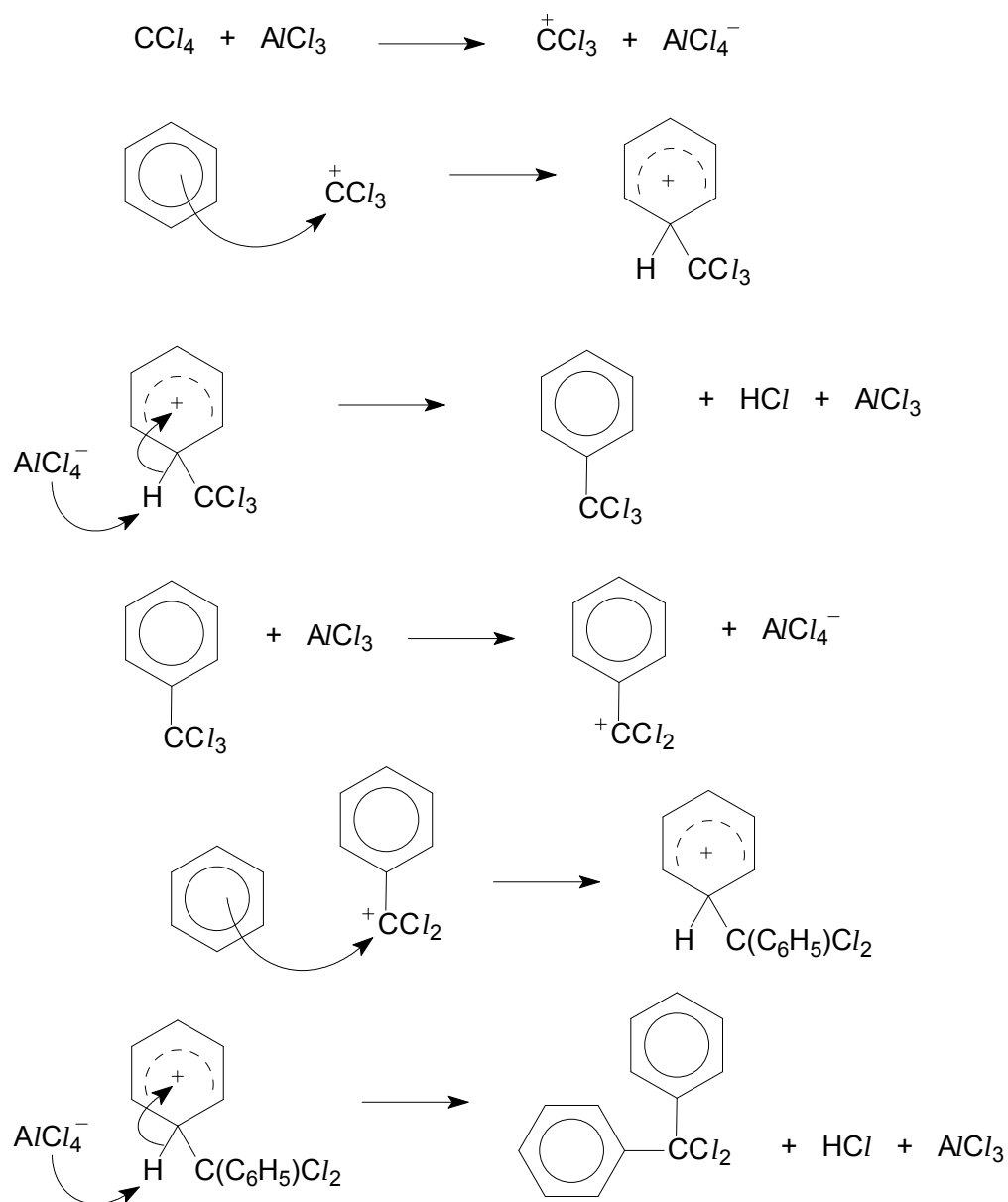
Universal Indicator solution turned red (~pH 1)

[1]: any 1 equation
[1]: observation

- (c) (i) AlCl_3 hydrolyses in water, resulting in absence of catalyst to generate electrophile for Friedel–craft alkylation to take place. [1]

(ii) electrophilic substitution

[4]



[1]: name of mechanism

(correct for at least one complete ES mechanism for each of the following)

[1]: charges and intermediates

[1]: lone pairs and curly arrows

[1]: generation of electrophile and regeneration of catalyst

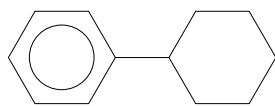
- (iii) The two large phenyl groups bonded to the small carbon atom repel the 3rd phenyl group from attacking the electrophilic C atom. (steric hindrance)

[1]

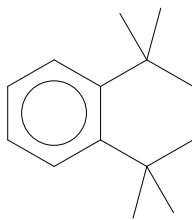
(accept stability of the electrophile generated due to resonance, resulting in larger E_a for further alkylation)

(iv)

[2]



A

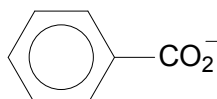
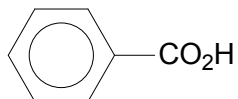


B

(v) KMnO_4 , dil H_2SO_4 / NaOH(aq) , heat

[2]

A will decolourise purple KMnO_4 . Purple KMnO_4 remains for **B**.



formed for **A**.

[1]: reagent and condition

[1]: structure of product

(vi) The triphenylmethyl cation generated is relatively unreactive / weak electrophile,
due to extensive resonance charge delocalisation,
and hence can only substitute highly activated aromatic rings like phenol.

[1]

[1]

- 3 (a) (i) from sp in CO_2 to sp² in RCO_2^- [1]
- (ii) acid-base reaction [1]
- (iii) $\text{CH}_3\text{CO}_2\text{H}$ [1]
- (iv) $\text{C}_4\text{H}_6\text{O}_3$ [1]

FYI

F is $(\text{CH}_3\text{CO})_2\text{O}$

- (b) (i) Intramolecular hydrogen bonding between $-\text{OH}$ and $-\text{CO}_2\text{H}$ groups, resulting in less energy required to break the less extensive intermolecular hydrogen bonding between salicylic acid molecules hence, lower m.p. [2]

(accept explanation in terms of better packing (denser) for 4-hydroxybenzoic acid due to its more linear shape and larger surface area for id-id interactions)

[1]: recognise intramolecular hydrogen bonding result in less extensive intermolecular hydrogen bonding

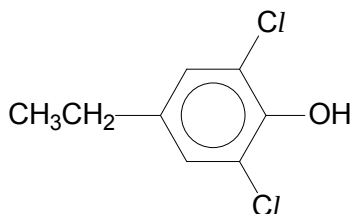
[1]: less energy to overcome IMF

- (ii) Salicylic acid has a lower pK_a , higher K_a / is a stronger acid than 4-hydroxybenzoic acid [2]
Intramolecular hydrogen bonding helps to disperse the negative charge on the conjugate base, stabilising it to a greater extent.

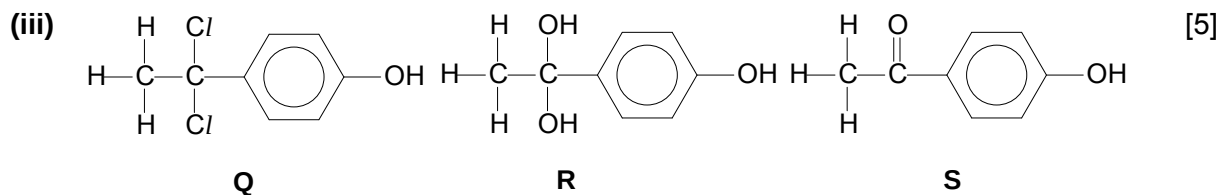
[1]: relationship between pK_{a1} and K_{a1} / strength of acid

[1]: recognise intramolecular hydrogen bonding in stabilising monoanion

- (c) (i) [1]



- (ii) P is more acidic [1]
 Presence of electron-withdrawing Cl groups help to disperse the negative charge of the conjugate base and stabilising it to a greater extent. [1]



[1] each correct structure

Q can undergo nucleophilic substitution with NaOH(aq) because the Cl atoms are bonded to sp^3 C atom / it is a chloroalkane. **R** is a diol.

R undergoes elimination to form **S**. **S** is an aldehyde / ketone.

The phenolic group in **S** reacts with NaOH(aq) (acid–base reaction) to form an ionic salt and hence is soluble.

S is a ketone and undergoes condensation reaction with 2,4–DNPH but does not react with Tollens' reagent because it is not an aldehyde.

S gives a yellow precipitate with $I_2(aq)/OH^-(aq)$ (oxidation) because it contains the CH_3COR structure.

OR

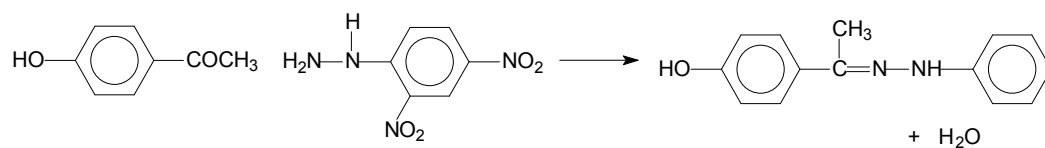
Observation	Type of reaction	Deduction
Compound Q reacts with NaOH(aq), followed by acidification, to give R , $C_8H_{10}O_3$.	nucleophilic substitution	Q is a chloroalkane / contains Cl atoms bonded to sp^3 C atom. R is a diol.
R immediately loses water to form S , $C_8H_8O_2$.	elimination (of H_2O)	S is an aldehyde / ketone.
Compound S is insoluble in water but it dissolves in NaOH(aq).	acid–base reaction	S contains phenol or carboxylic acid group.
S reacts with 2,4–DNPH but not with Tollens' reagent.	condensation	S contains ketone group.
S reacts with alkaline aqueous iodine	oxidation	S contains $RCOCH_3$.

[2]: 8–11 points

[1]: 4–8 points

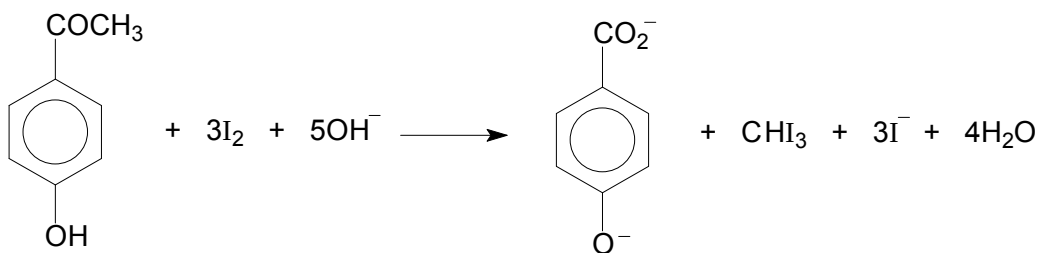
(iv) between S and 2,4-dinitrophenylhydrazine

[1]



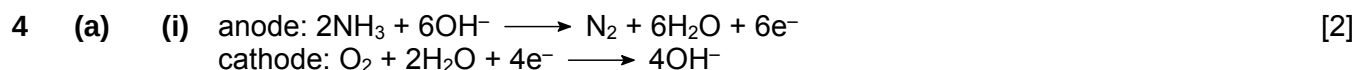
between S and alkaline aqueous iodine

[1]

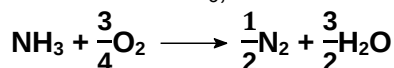


(v) presence of light / uv light

[1]



for 1 mol of NH_3 ,



(iii) $\Delta G^\ominus = -nFE_{\text{cell}}^\ominus$ [2]
 $-339 \times 1000 = -(3)(96500)E_{\text{cell}}^\ominus$
 $E_{\text{cell}}^\ominus = \underline{+1.17 \text{ V}}$

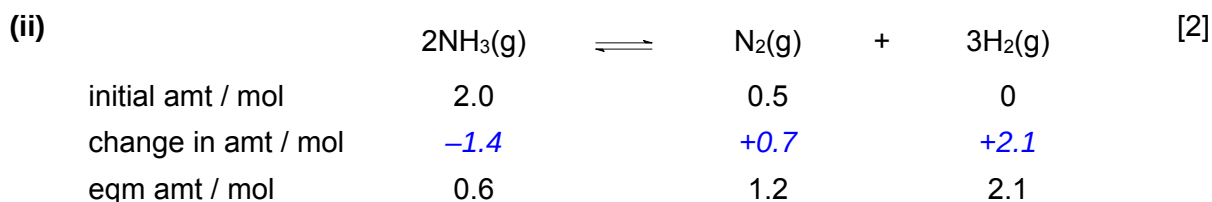
(iv) $E_{\text{cell}}^\ominus = E_{\text{O}_2/\text{OH}^-}^\ominus - E_{\text{N}_2/\text{NH}_3}^\ominus$ [1]
 $+1.17 = 0.40 - E_{\text{N}_2/\text{NH}_3}^\ominus$
 $E_{\text{N}_2/\text{NH}_3}^\ominus = \underline{-0.77 \text{ V}}$

(v) Ammonia is a weak base, will react readily with the acidic membrane to form salts, which increases the resistance of the cell rapidly. [1]

(b) (i) $K_c = \frac{[\text{N}_2][\text{H}_2]^3}{[\text{NH}_3]^2} \text{ mol}^2 \text{ dm}^{-6}$ [2]

[1]: correct expression

[1]: correct units



$$[\text{NH}_3] = \frac{0.6}{2} = \underline{0.3 \text{ mol dm}^{-3}}$$

$$[\text{N}_2] = \frac{1.2}{2} = \underline{0.6 \text{ mol dm}^{-3}}$$

$$[\text{H}_2] = \frac{2.1}{2} = \underline{1.05 \text{ mol dm}^{-3}}$$

[2]: all 3 concentrations are correct

[1] any one concentration

(iii) $K_c = \frac{[\text{N}_2][\text{H}_2]^3}{[\text{NH}_3]^2}$ [1]
 $= \frac{0.6(1.05)^3}{(0.3)^2}$
 $= \underline{7.72 \text{ mol}^2 \text{ dm}^{-6}}$ ecf

(can give credit for correct units if missing in (i))

(iv) $\Delta G^\ominus = -(8.31)(670) \times \ln(7.72)$ [1]
 $= \underline{-11400 \text{ J mol}^{-1}}$

(v) $\Delta G^\ominus = \Delta H^\ominus - T\Delta S^\ominus$ [2]
 $-11.4 = +92 - (670)\Delta S^\ominus$
 $\Delta S^\ominus = \frac{+0.154 \text{ kJ mol}^{-1} \text{ K}^{-1}}{= +154 \text{ J mol}^{-1} \text{ K}^{-1}}$

Entropy change for the reaction is positive, indicating an increase in ways of distributing energy as seen from the equation given where there is an increase in number of moles of gaseous particles.

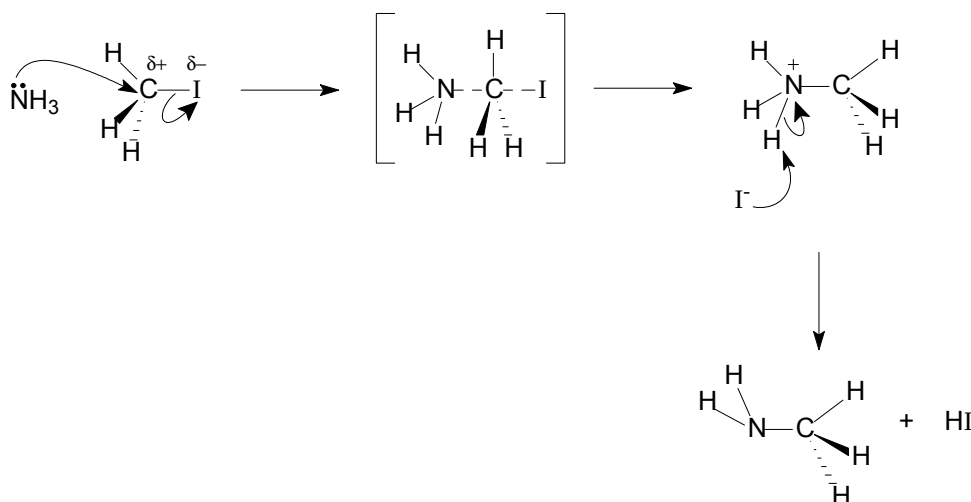
[1]: $\Delta S^\ominus$, include units

[1]: explaining link between entropy change and disorder of reaction

(vi) Partial pressures of all gases decrease, [1]
 Reaction results in greater number of moles of gaseous products and hence, a greater decrease in partial pressure on the products side,
 the position of equilibrium shifts right to minimise the increased in volume.

However, as there is no change in temperature, K_p remains constant. [1]

(c) Bimolecular nucleophilic substitution (S_N2) [3]



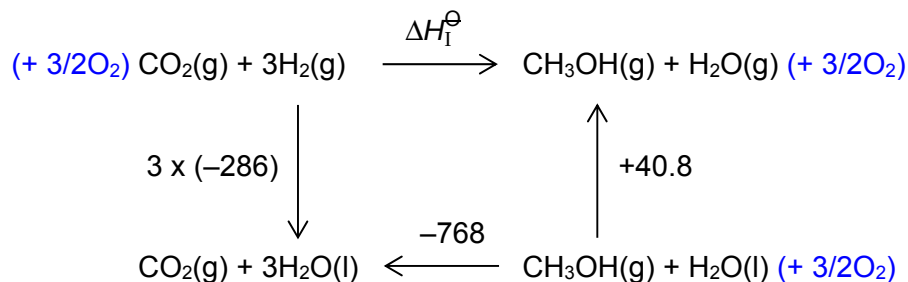
[1]: type of reaction

[1]: correct dipoles and lone pair AND curly arrows to show all the relevant bond breaking/formation (ignore if missing for the last step)

[1]: correct structures for transition state, positively charged amine salt and final product

5 (a)

[3]



By Hess' Law,

$$\begin{aligned}
 \Delta H_f^\ominus &= 3(-286) - (-768) + 40.8 \\
 &= \underline{-49.2 \text{ kJ mol}^{-1}}
 \end{aligned}$$

[1]: correct conversion with correct state symbols (ignore balancing equation)

[1]: correct enthalpy change values (i.e. multiply with correct factors)

[1]: correct answer

- (b) (i) anode: $\text{CH}_3\text{OH} + \text{H}_2\text{O} \longrightarrow \text{CO}_2 + 6\text{H}^+ + 6\text{e}^-$ [2]
 cathode: $\text{O}_2 + 4\text{H}^+ + 4\text{e}^- \longrightarrow 2\text{H}_2\text{O}$

[1] each correct half-equations

- (ii) Alkaline electrolyte reacts with (acidic) carbon dioxide produced to form an insoluble carbonate which degrades the fuel cell and leads to poor performance. [1]

- (iii) $\Delta G^\ominus = -nFE_{\text{cell}}^\ominus$ [1]
 $= -(6)(96500)(1.18)$
 $= \underline{-683\,000 \text{ J mol}^{-1}}$

- (iv) $\text{CH}_3\text{OH}(\text{l}) + \frac{3}{2} \text{O}_2(\text{g}) \longrightarrow \text{CO}_2(\text{g}) + 2\text{H}_2\text{O}(\text{l})$ [1]

$$\begin{aligned}
 \Delta H_r^\ominus &= [(-394) + 2(-286)] - (-238) \\
 &= \underline{-728 \text{ kJ mol}^{-1}}
 \end{aligned}$$

- (v) $\Delta G_r^\ominus = \Delta H_r^\ominus - T\Delta S_r^\ominus$ [1]
 $(-683) = (-728) - 298\Delta S_r^\ominus$
 $\Delta S_r^\ominus = -0.151 \text{ kJ mol}^{-1} \text{ K}^{-1}$
 $= \underline{-151 \text{ J mol}^{-1} \text{ K}^{-1}}$

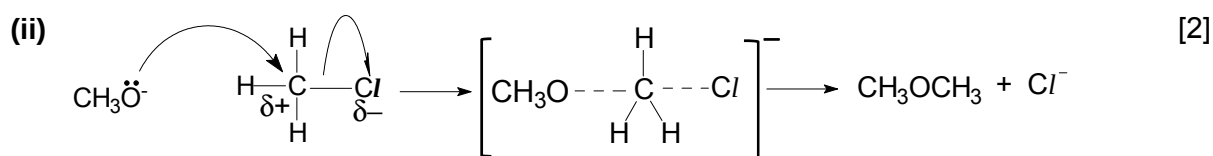
- (vi) At 25 °C, $\text{CH}_3\text{OH}(\text{l}) + \frac{3}{2} \text{O}_2(\text{g}) \longrightarrow \text{CO}_2(\text{g}) + 2\text{H}_2\text{O}(\text{l})$ [2]
 At 80 °C, $\text{CH}_3\text{OH}(\text{g}) + \frac{3}{2} \text{O}_2(\text{g}) \longrightarrow \text{CO}_2(\text{g}) + 2\text{H}_2\text{O}(\text{l})$

At 80 °C, methanol exists in the gaseous state.

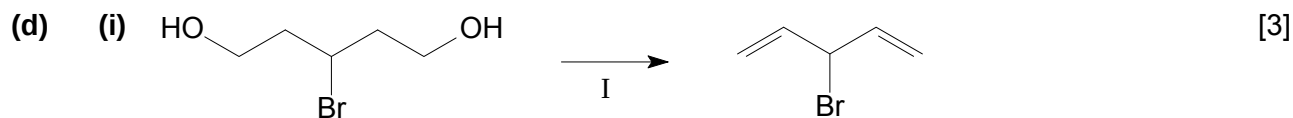
Hence, when operating temperature 80 °C, there is a larger decrease in number gaseous molecules in the cell reaction, resulting in greater decrease in number of ways in distributing the energy. Hence ΔS_r is more negative than $\Delta S_r^\ominus$.

[1]: recognise that methanol exists in the gaseous state at 80 °C and entropy change will be more negative

[1]: explain it is due to larger decrease in number of gaseous particles



[1]: partial charges indicated + arrows correctly show electron pair on nucleophile attacking δ^+ of C-Cl bond and electrons move to Cl
 [1]: correct transition state



I: excess conc. H_2SO_4 , heat (under reflux) / 180°C

II: KMnO_4 , NaOH(aq) , heat (under reflux)

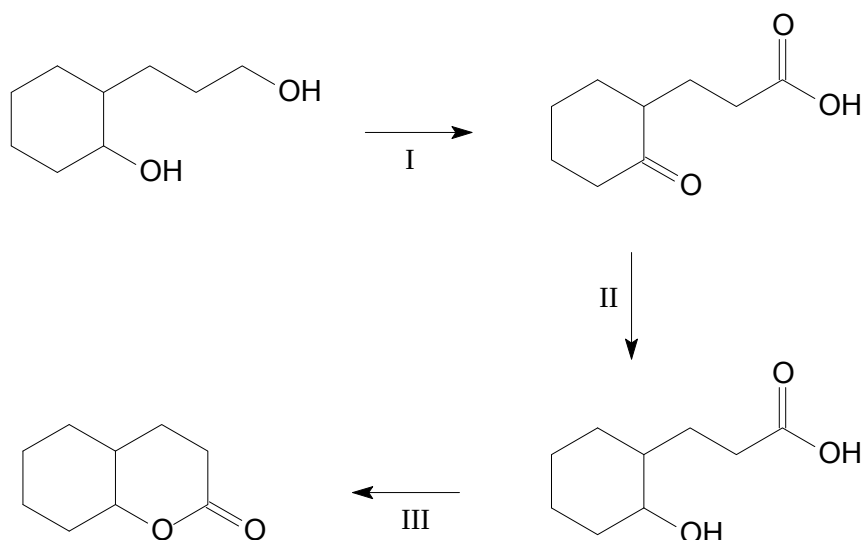
III: dilute H_2SO_4 / dilute HCl

[1]: correct reagent and condition for step 1 and its corresponding product
 [1]: correct reagent and condition for step 2 and its corresponding product
 [1]: correct reagent and condition for step 3

Do not award if Step II uses acidified KMnO_4 , followed by NaOH(aq) / KOH(aq) with heat in Step III. This will need a Step IV to acidify to liberate the free acid.

(ii)

[3]



I: KMnO_4 , dilute H_2SO_4 , heat (under reflux)

II: NaBH_4 (in ethanol) // H_2 , Pt / Pd, r.t. // H_2 , Ni, heat

III: conc. H_2SO_4 , heat (under reflux)

[1]: correct reagent and condition for step 1 and its corresponding product

[1]: correct reagent and condition for step 2 and its corresponding product

[1]: correct reagent and condition for step 3